Start

Input Data (Real-time & Forecasted)

- Load profile (All India / IIT Kanpur)
- Solar PV generation (Kaggle)
- Electricity price (IEX / Mendeley)
- EV data (arrival, departure, SOC, battery, max power)
- EVCS / Port data (stations, ports, power limits)
- System parameters (grid/solar capacity, efficiency, etc.)

Data Sources
- Load: All India / IIT Kanpur
- Solar: Kaggle
- Price: IEX / Mendeley
- EVCS: Project Data / Assumptions

Data Loaded Successfully?

No → **Show Error / Missing File Message** → (back to Input Data)

Yes

Data Preprocessing

- Convert to 24-hour horizon (96 time slots)
- Prepare matrices & parameters
- Format data (load, solar, price, EV, EVCS)

Processed Data
- Load profile (96 × n)
- Solar generation (96 × 1)
- Price (96 × 1)
- EV / EVCS data (structured)

Create Decision Variables (MILP)

x(e,p,t)	→	EV-port assignment (binary)
P(e,p,t)	→	Charging power (continuous)
Ppv(e,p,t)	→	Solar power (continuous)
Pg(e,p,t)	→	Grid power (continuous)
SOC(e,t)	→	State of Charge (continuous)

Variable Types
- x : binary
- P, Ppv, Pg, SOC : continuous

MILP Constraints (Core Equations)

1. EV Assignment:

$$\sum_{p} x_{e,p,t} \le 1$$

2. Charging Power Limit:

$$P_{e,p,t} \le P_{e,p}^{\max} \cdot x_{e,p,t}$$

3. Station Capacity:

$$\sum_{e,p \in s} P_{e,p,t} \le P_{s}^{\text{station}}$$

4. Power Balance:

$$P_{e,p,t} = P_{e,p,t}^{\text{PV}} + P_{e,p,t}^{\text{G}}$$

5. Solar Limit:

$$P_{e,p,t}^{\text{PV}} \le P_{t}^{\text{PV,max}}$$

6. Grid Limit:

$$P_{e,p,t}^{\text{G}} \le P_{t}^{\text{G,max}}$$

7. SOC Dynamics:

$$SOC_{e,t+1} = SOC_{e,t} + \eta \cdot P_{e,t} \cdot \frac{\Delta t}{E_e}$$

8. Required SOC at Departure:

$$SOC_{e,D_e} \ge SOC_{e}^{\text{req}}$$

Key Parameters
- η : charging efficiency
- Δt : 15 min (0.25 h)
- E_e : battery capacity
- $P_{e,p}^{\max}$: port max power
- P_{s}^{station} : station capacity
- $P_{t}^{\text{PV,max}}$, $P_{t}^{\text{G,max}}$: limits

Construct Objective Function (Profit Maximization)

System Profit (40%)
- Charging Revenue
- – Grid Energy Cost
- – Operating Cost

Participant Profit (40%)
- Charging Revenue/ Benefit
- – Charging Cost

Solar Utilization (20%)
- Solar Energy Contribution

Maximize $Z = w_1 P_{\text{system}} + w_2 P_{\text{participant}} + w_3 E_{\text{solar}}$

($w_1 = 0.40$, $w_2 = 0.40$, $w_3 = 0.20$)

Objective Components

P_{system} = Revenue − Grid Cost − Op. Cost

$P_{\text{participant}}$ = Revenue − Charging Cost

$E_{\text{solar}} = \sum P_{e,p,t}^{\text{PV}} \cdot \Delta t$

For MATLAB (minimization)

min −Z

(or equivalently minimize −Z)

Convert to MATLAB MILP Form

$$\begin{aligned} \min \quad & f^T x \\ \text{subject to} \quad & Ax \le b \\ & Aeqx = beq \\ & lb \le x \le ub \\ & \text{intcon (integer variables)} \end{aligned}$$

MATLAB Structures
- f : objective coefficients
- A, b : inequality constraints
- Aeq, beq : equality constraints
- lb, ub : bounds
- intcon : integer variables

Call intlinprog (MILP Solver)

Search for feasible & optimal solution

Solver Process
- Feasible solution search
- Constraint checking
- Objective evaluation
- Optimal solution (xsol)

Optimal Solution Found?

No → **Adjust Weights / Parameters (or check constraints)** → (back to Convert to MATLAB MILP Form)

Yes

Extract Solution (xsol)

- EV-port assignment
- Charging power (P)
- Solar power (Ppv)
- Grid power (Pg)
- SOC profile

Validate Solution

SOC requirement met? → Yes → Power limits met? → Yes → Station capacity met? → Yes → Energy balance met? → Yes

No → **Re-run Optimization (Adjust Weights / Parameters)** → (back to Convert to MATLAB MILP Form)

Generate Results & Graphs

- EV charging schedule
- EV-port allocation
- Solar / Grid allocation
- GOC profiles
- Station utilization
- Energy & economic metrics

Output
- Optimal charging schedule
- Energy distribution
- Profit / cost details
- Graphs (plots)

End